PS#13 - Taylor series - Answer key

1. (AC Activity 8a.4.2) Consider the infinite Taylor series given by $T(x) = \sum_{k=1}^{\infty} \frac{1}{k \cdot 2^k}(x-1)^k$.

   (a) As described in the statment of the Ratio Test, let $r_n(x)$ be the ratio of the $(n+1)$st term of $T(x)$ to the nth term of $T(x)$. Find the simplest formula that you can for $r_n(x)$.

   $r_n(x) = \frac{n}{n+1} \cdot \frac{1}{2} \cdot (x-1)$.

   (b) Let $r(x) = \lim_{n\to\infty} r_n(x)$. Evaluate this limit to find the simplest formula you can for $r(x)$.

   $r(x) = \frac{1}{2}(x-1)$.

   (c) For what values of x is $|r(x)| < 1$? $-1 < x < 3$.

2. In this problem we're going to explore the Taylor series for $f(x) = \sqrt{x}$ centered at $a = 1$.

   (a) Find the Taylor polynomials for $f(x) = \sqrt{x}$ up through order 5, centered at $a = 1$.

   $T_5(x) = 1 + \frac{1}{2}(x-1) - \frac{1}{2! \cdot 2^2}(x-1)^2 + \frac{1 \cdot 3}{3! \cdot 2^3}(x-1)^3 - \frac{1 \cdot 3 \cdot 5}{4! \cdot 2^4}(x-1)^4 + \frac{1 \cdot 3 \cdot 5 \cdot 7}{5! \cdot 2^5}(x-1)^5$.

   (b) Spot the pattern in the coefficients.

   $c_k = (-1)^{k+1} \frac{1 \cdot 3 \cdot \ldots \cdot (2k-3)}{k! \cdot 2^k}$.

   (c) Use the pattern you found to do the ratio test.

   $$|r_n(x)| = \frac{(2n-1)}{(n+1) \cdot 2} \cdot |x-1|$$
   $$|r(x)| = |x-1| \lim_{n\to\infty} \frac{2n-1}{2n+2} = |x-1| \cdot 1$$

   (d) What is the "interval of convergence" – the x-values for which the Taylor series will converge? Why does your answer make sense based on the graph of the original function $f(x) = \sqrt{x}$?

   This series will converge whenever $|r(x)| = |x-1| < 1$ – that is, for $0 < x < 2$. This makes sense because the graph of $\sqrt{x}$ stops at $x = 0$.

   (e) Use $T_5(x)$ to find a good fraction approximation for $\sqrt{2} \approx 1.414\ldots$

   $$\begin{aligned} T_5(2) &= 1 + \frac{1}{2}(2-1) - \frac{1}{2! \cdot 2^2}(2-1)^2 + \frac{1 \cdot 3}{3! \cdot 2^3}(2-1)^3 - \frac{1 \cdot 3 \cdot 5}{4! \cdot 2^4}(2-1)^4 + \frac{1 \cdot 3 \cdot 5 \cdot 7}{5! \cdot 2^5}(2-1)^5 \\ &= 1 + \frac{1}{2} - \frac{1}{2! \cdot 2^2} + \frac{1 \cdot 3}{3! \cdot 2^3} - \frac{1 \cdot 3 \cdot 5}{4! \cdot 2^4} + \frac{1 \cdot 3 \cdot 5 \cdot 7}{5! \cdot 2^5} \\ &= \frac{365}{256} = 1.42578125. \end{aligned}$$

   Not bad!